

1. Administrative & Study Identification

Full Study Title: An Open-Label, Multicenter, Phase I Study Evaluating the Safety and Pharmacokinetics of Escalating Doses of Forimtamig (R07425781) in Participants With Relapsed or Refractory Multiple Myeloma **Short Title/Acronym:** BP42233 **Protocol Number:** BP42233 (EudraCT: 2020-002012-46) **NCT Number:** NCT04557150 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** Forimtamig (R07425781 / RG6234) - a GPRC5DxCD3 bispecific antibody **Phase of Development:** Phase 1 **Medical Monitor:** Hoffmann-La Roche Clinical Study Team / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective: To evaluate the safety, tolerability, and pharmacokinetics of escalating doses of forimtamig (R07425781) and to determine the Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D) in participants with relapsed or refractory multiple myeloma (r/r MM). **Secondary Objectives:** To evaluate preliminary anti-myeloma clinical activity including Overall Response Rate (ORR), Very Good Partial Response (VGPR), Complete Response (CR), Progression-Free Survival (PFS), Duration of Response (DOR), as well as pharmacodynamics and immunogenicity.

2.2 Background & Rationale To address the critical unmet medical need in patients with relapsed or refractory multiple myeloma (r/r MM) who have exhausted standard-of-care treatments. Although there are many treatment options for people diagnosed with MM, the cancer often returns or becomes refractory to immunomodulatory drugs and proteasome inhibitors. Forimtamig is a novel "off-the-shelf" T-cell bispecific antibody with a 2+1 format that binds to CD3 on T-cells and GPRC5D on myeloma plasma cells, bringing them together to induce rapid T-cell-mediated immune destruction of the myeloma cells.

3. Study Design & Methodology

3.1 Design Framework

Study Type: Interventional **Control Type:** Uncontrolled (Single-arm during Dose Escalation) **Blinding:** Open-label **Randomization:** N/A for Part 1 (Dose Escalation); Randomized for Part 2 (Dose Expansion)

3.2 Planned Enrollment & Duration

Target \$N\$: 225 participants (Actual Enrollment) **Number of Sites:** Approximately 20 locations globally **Estimated Study Start:** 11-Nov-2020 **Estimated Primary Completion:** 30-Mar-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female participants, Age 18 to 100 years. 2. Previously diagnosed with Multiple Myeloma (MM) based on standard criteria, with measurable disease. 3. Participants with r/r MM who have previously received therapy with an Immunomodulatory drug (IMiD) and Proteasome Inhibitor (PI) and are intolerant to or have no other option for standard-of-care treatment. 4. Life expectancy of at least 12 weeks. 5. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 6. Agreement to provide protocol-specific biopsy material.

4.2 Exclusion Criteria 1. Inability to comply with protocol-mandated hospitalization and activities restrictions. 2. Pregnant or breastfeeding or intending to become pregnant during the study or within 3 months after last dose of study drug. 3. Autologous or allogeneic stem cell transplantation (SCT) within 100 days prior to first forimtamig infusion and/or signs of chronic graft versus host disease. 4. Prior use of any monoclonal antibody, radioimmunoconjugate, or antibody-drug conjugate for MM treatment within 2 weeks before first forimtamig administration. 5. Prior treatment with systemic immunotherapeutic agents within 2 weeks before first forimtamig administration. 6. Active auto-immune disease or flare within 6 months prior to start of study treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen

Dosage/Frequency: Escalating doses of forimtamig utilizing a step-up dosing regimen (e.g., step-up doses on Day 1 and Day 8, with the first full target dose given on Day 15) followed by administration every 2 weeks. Target optimal doses vary based on cohort assignment. **Route of Administration:** Intravenous (IV) infusion or Subcutaneous (SC) injection. **Duration of Treatment:** Up to 1 year, or up to 2 years if the participant continues to derive clinical benefit.

5.2 Concomitant & Prohibited Therapy

Permitted: Corticosteroid premedications and step-up dosing to mitigate the risk of Cytokine Release Syndrome (CRS) during Cycle 1; Tocilizumab for the management of CRS. **Prohibited:** Concurrent anti-cancer therapy, systemic immunotherapeutic agents,

radiotherapy (except limited palliative field for bone pain), and other investigational drugs.

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures
Screening	2 to 4 weeks prior	Informed Consent, Medical History, Baseline disease assessment, Biopsy
Baseline/Cycle 1	Days 1, 8, 15	Step-up dosing administration; Hospitalization for overnight safety monitoring to manage potential CRS
Interim	Every 2 weeks	Ongoing forintamig administration, Safety Labs, Efficacy Assessment, AE tracking
End of Study	Up to Year 1-2	Final clinical response evaluation, Exit Interview, Transition to follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability, defined by the incidence and severity of Adverse Events (AEs) including Dose-Limiting Toxicities (DLTs), to determine the Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D).

Measurement Method: Clinical observation and laboratory evaluations graded according to standard toxicity criteria (CTCAE) during the dose escalation phase.

7.2 Secondary & Exploratory Endpoints 1. Overall Response Rate (ORR), Very Good Partial Response (VGPR) rate, and Complete Response (CR) rate. 2. Duration of Response (DOR) and Progression-Free Survival (PFS). 3. Pharmacokinetics (Cmax, AUC) and immunogenicity profiles.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Intensive monitoring for Cytokine Release Syndrome (CRS), skin and mucosal toxicity, cytopenias, and infections. Monitored through frequent clinical visits and lab assessments.

Serious Adverse Events (SAEs): Inpatient hospitalization mandated during initial doses specifically for strict observation and rapid management of severe CRS or ICANS.

Data Monitoring Committee (DMC): Internal Safety Review Committee responsible for analyzing DLTs and directing dose escalation/expansion decisions.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set (all enrolled participants who receive at least one dose of study drug) and Efficacy Analysis Set. **Sample Size Justification:** Target

enrollment of \$N=225\$ appropriately accommodates a standard phase 1 dose-escalation statistical design (e.g., 3+3 or continual reassessment method) followed by randomized dose-expansion cohorts to ensure robust initial efficacy and safety estimates. **Handling of Missing Data:** Standard Phase I oncology conventions; patients dropping out prior to DLT evaluation windows without experiencing a DLT may be replaced to ensure adequate cohort sizes. Time-to-event variables (PFS/DOR) handled via right-censoring.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine rigorous onsite and remote monitoring of investigator sites; strict adherence tracking for AE/SAE reporting timelines. **Audit Trail:** Electronic Case Report Form (eCRF) source data verification and comprehensive data management locking procedures.

11. Study Identifiers & Governance

Primary ID: BP42233 **Registry Number:** NCT04557150 **Secondary IDs:** 2020-002012-46, 2023-504571-25-00, 21030043 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** A Study Evaluating the Safety and Pharmacokinetics of Escalating Doses of Forimtamig in Participants With Relapsed or Refractory Multiple Myeloma (r/r MM) **Official Regulatory Title:** An Open-Label, Multicenter, Phase I Study Evaluating the Safety and Pharmacokinetics of Escalating Doses of Forimtamig (R07425781) in Participants With Relapsed or Refractory Multiple Myeloma

12. Clinical Framework (Multiple Myeloma Specific)

Primary Condition: Relapsed or Refractory Multiple Myeloma (RRMM) **Diagnosis Threshold:** Heavily pretreated MM lacking standard-of-care options (prior IMiD and PI failure) **Medical Methodology:** Bispecific T-cell Engager Immunotherapy **Optimization Target:** Rapid T-cell activation and tumor cell killing via dual CD3 and GPRC5D targeting

13. Study Procedures & Methodology

Management Pathway: Step-up dosing with premedication to mitigate CRS risk. **Education Protocol (HCP):** Site training on specialized monitoring and early intervention protocols for CRS and neurologic toxicities. **Education Protocol (Patient):** Counseling on the requirements of hospitalization during initial dosing, AE

reporting (specifically fever or neurologic symptoms), informed consent, medical history collection, baseline disease assessment, and contraception adherence. **Quality Audit Mechanism:** Real-time safety data reviews by the medical monitor before escalating to subsequent dose cohorts.

14. Observation & Follow-Up Schedule

Total Duration: Up to 1-2 years of treatment, followed by post-treatment survival follow-up and an exit interview to transition to standard care. **Onsite Visit Cadence:** Intensive (several times a week) during Cycle 1; transitioning to once every 2 weeks during the maintenance phase. **Remote Monitoring Frequency:** Standard clinical trial frequency, adjusted dynamically during dose escalation. **Checklist Review Interval:** Routine safety, pharmacokinetic, and clinical response assessments evaluated cyclically.

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator				[Insert Name]			____		[DD-MMM-YYYY]							
	Clinical Operations Lead				[Insert Name]			____		[DD-MMM-YYYY]							
	Lead Statistician				[Insert Name]			____		[DD-MMM-YYYY]							